Comp-4400: Assignment 1: λ calculus (3 points)

Last Name	First Name	ID	Email ID	Signature	Marks	Marked by
				Lucas Sorel		

1. (0.5 point) Whether the following are valid λ expressions? Fill in the box with YES or NO.

$\lambda x.(fxy)a$	$\lambda x.fxya$	$\lambda x.y$	$\lambda x\ y$
$\lambda x.(\lambda f.xy)a$	$\lambda z.fxya$	$y.\lambda x$	λx

2. (0.5 point) Identify the free and bound variables in the lambda expression: $\lambda x.\lambda y.x\ y\ z$.

   Free variables: Bound variables:

3. (1 point) Perform β reduction on the following λ expressions and write the result in the text box. If there are multiple β reduction steps, carry out all the reductions and write the final result; if there is no reduction can be performed, write "not reducible".

λ expression	Ressult
$\lambda x.(fxy)a$	
$\lambda x.(\lambda y.yfx)ab$	
$(\lambda x.\lambda y.yfx)ab$	

4. (1 Point) Given the following encoding for boolean logic. Prove that `or true false = true`. Write all the detailed steps.

$$\begin{aligned} true &= \lambda x.\lambda y.x \\ false &= \lambda x.\lambda y.y \\ or &= \lambda x.\lambda y.x\ true\ y \end{aligned}$$